

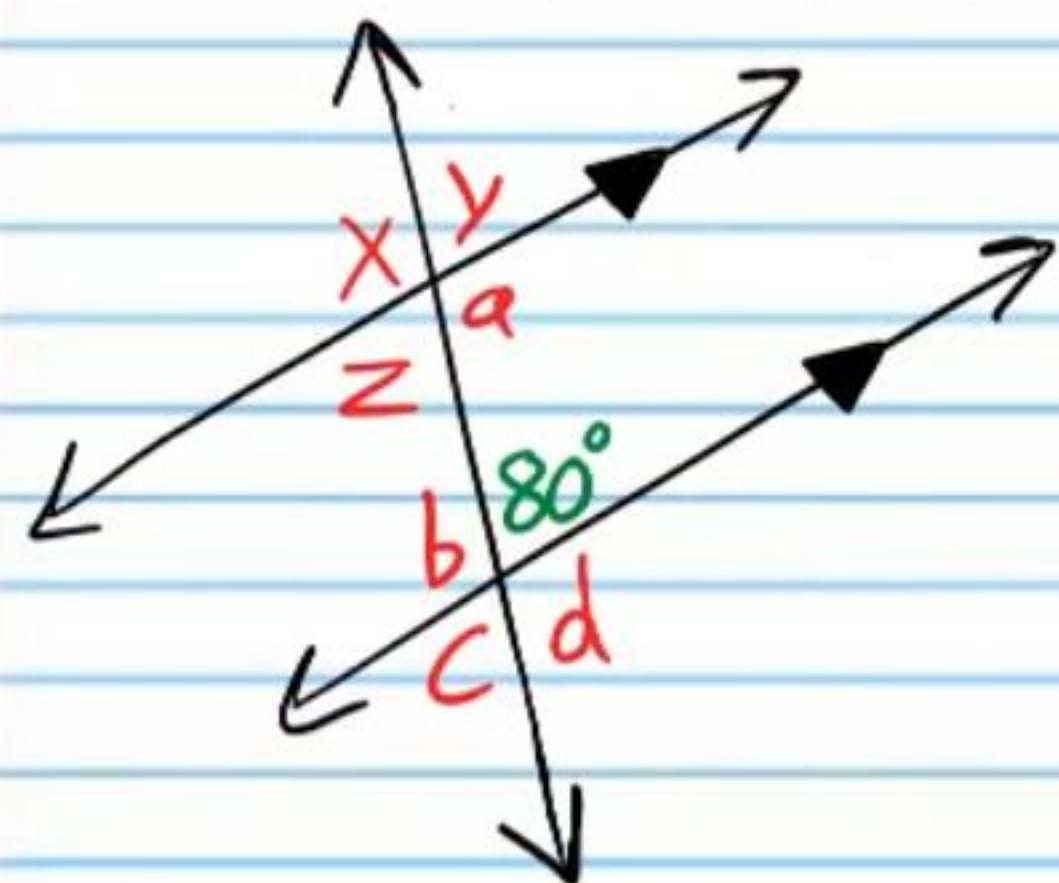
Geometry Test II

Name
Date
Course

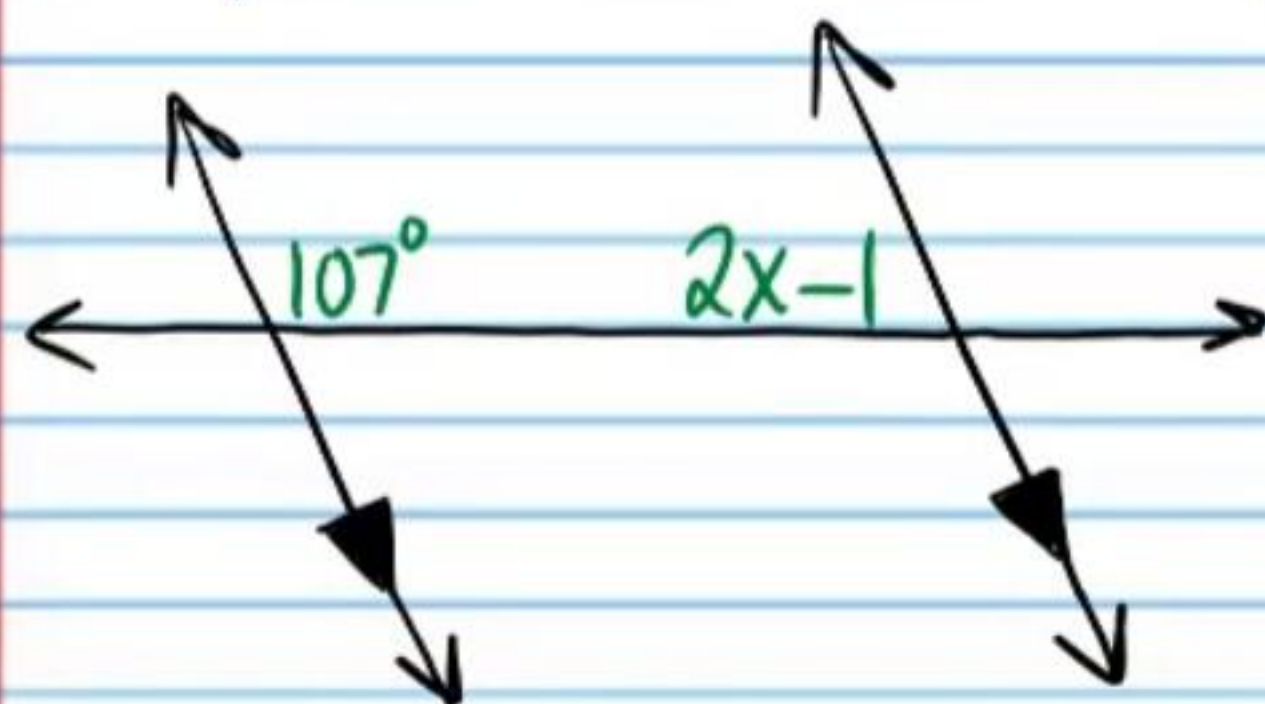
Instructions

- I) Write your name, the date, and the course title (i.e. Geometry).
- II) Show work when necessary.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) Each problem is worth four points for a total of 100 points.

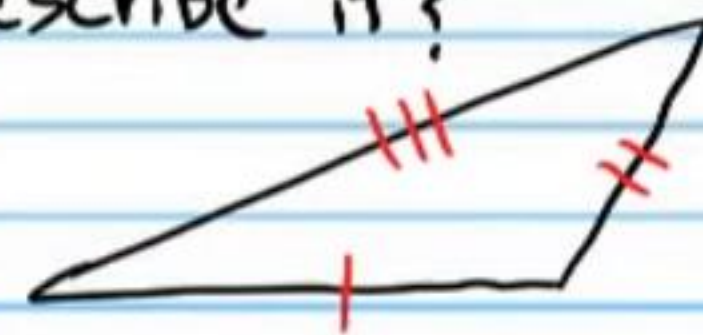
① Find $x, y, z, a, b, c,$ and d .



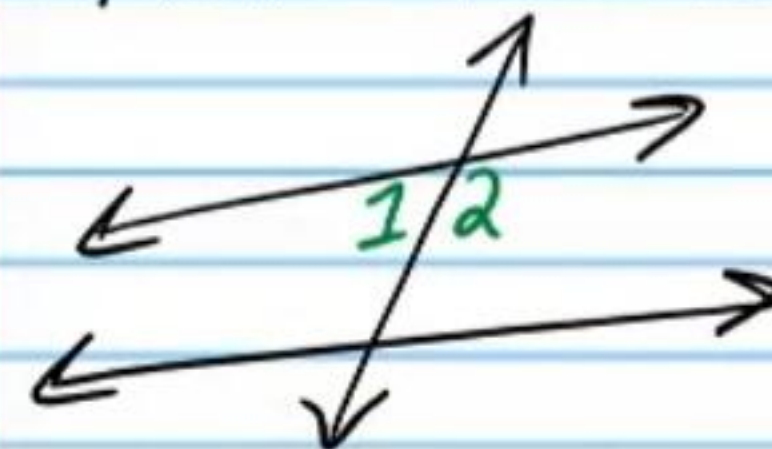
② Find x and state the theorem or postulate used. You do not have to write a two-column proof but you must show the equation used to find x .



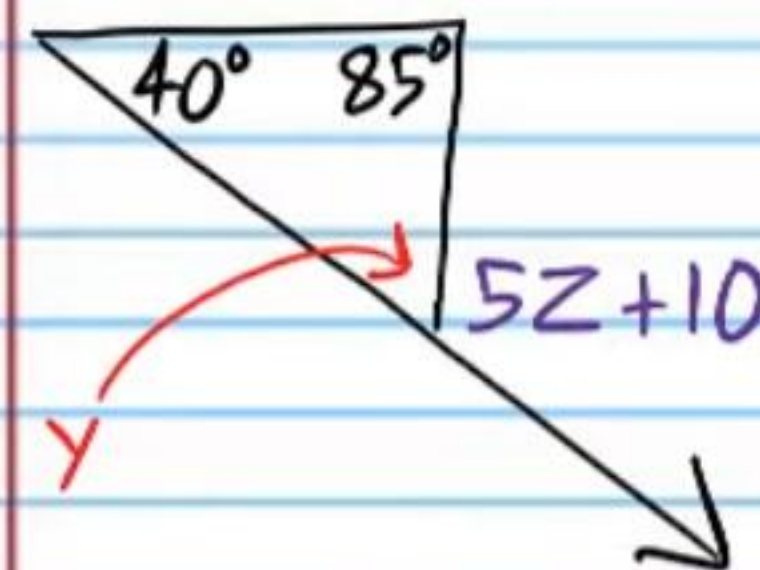
③ i) If the following triangle were classified by its sides, what word would be used to describe it?



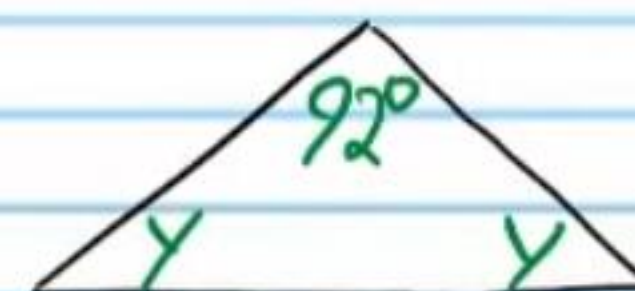
ii) Name the following angle pair.



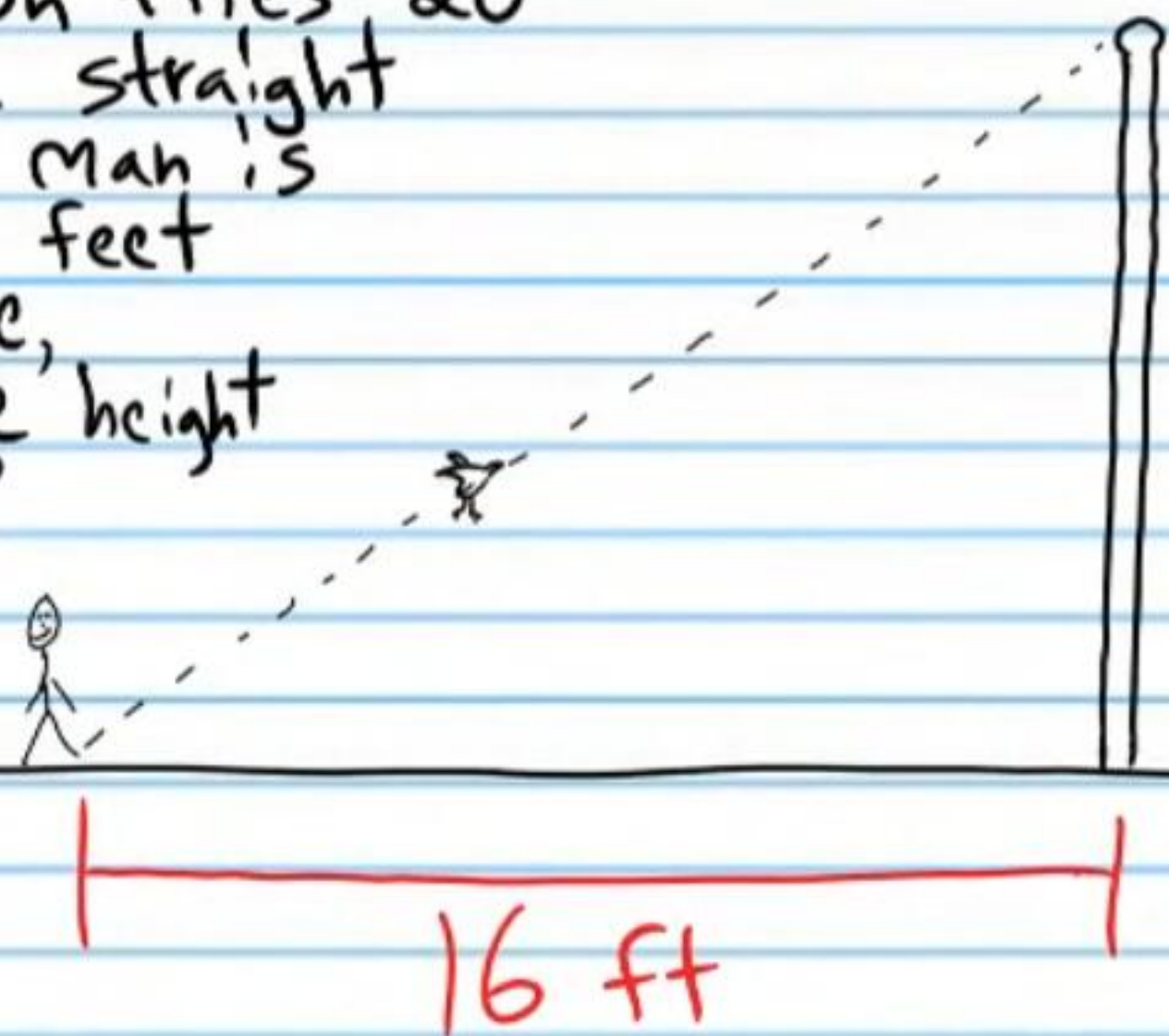
④ Find z .



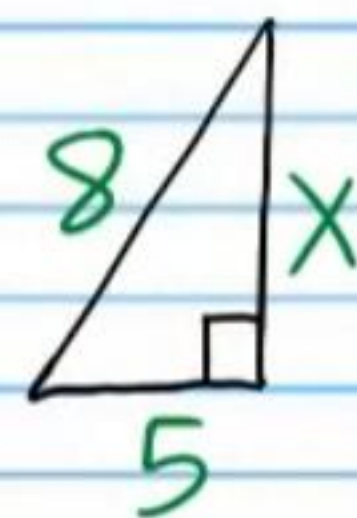
⑤ Find y .



⑥ A man lets his pigeon fly to the top of a flagpole. If the pigeon flies 20 feet along a straight path, and the man is standing 16 feet from the pole, what is the height of the pole?



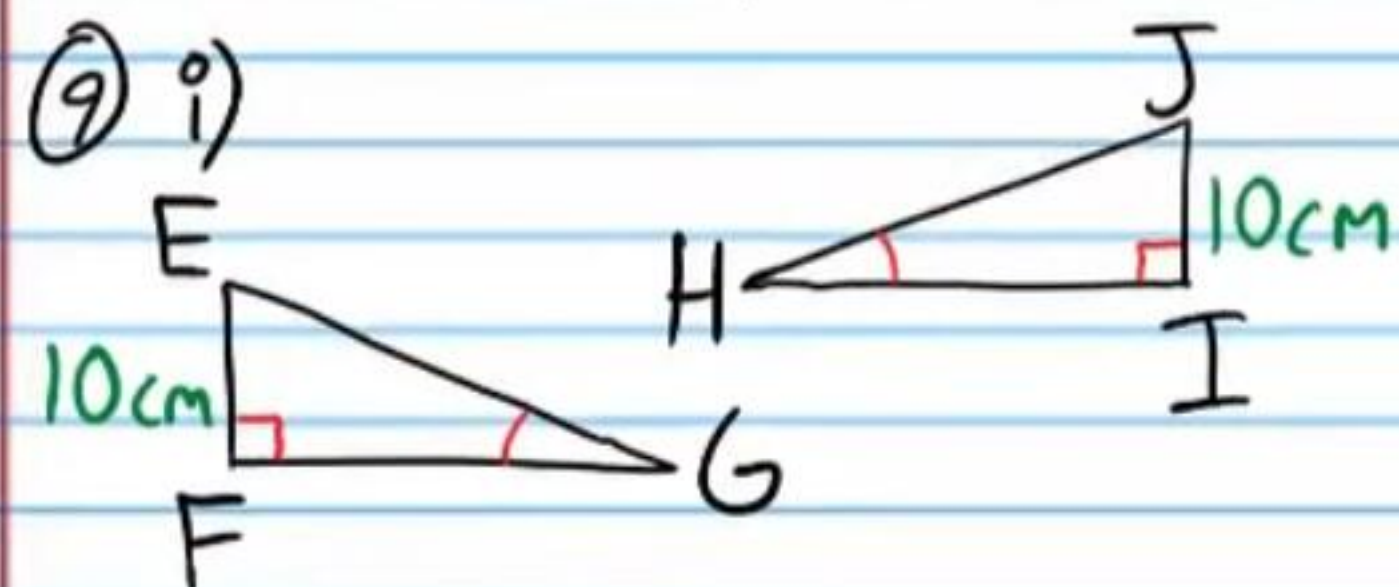
⑦ Find x .



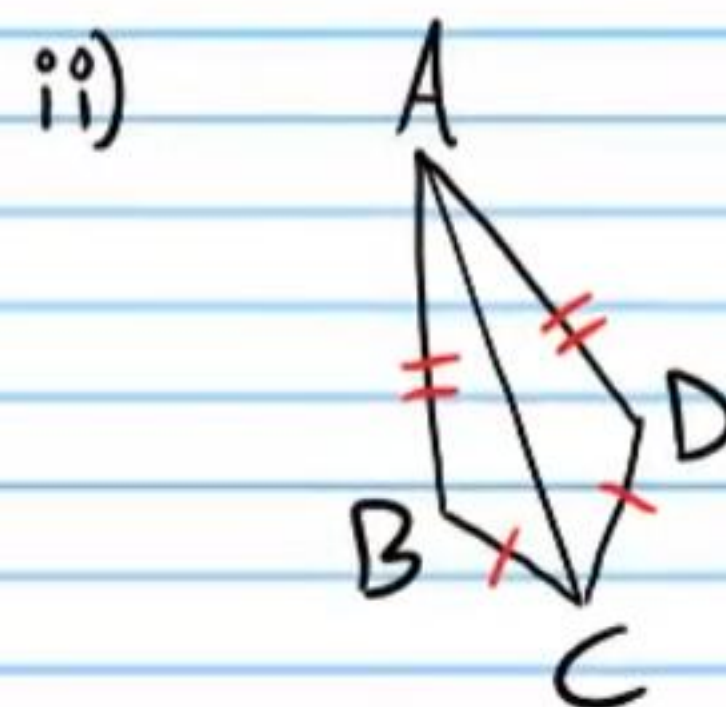
⑧ Find a .



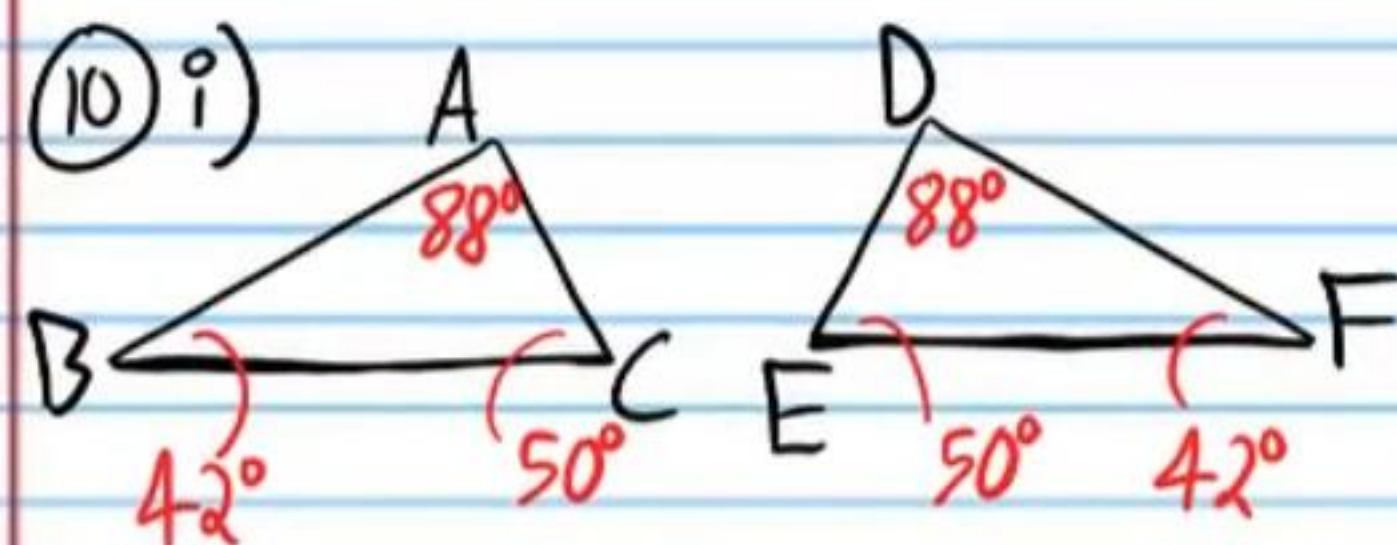
In problem #'s 9-11 below, determine if the triangles are congruent. If they are congruent, write a proper congruence statement and the triangle postulate or theorem that proves that they are congruent. You do not have to write a formal proof (i.e. a two-column proof).



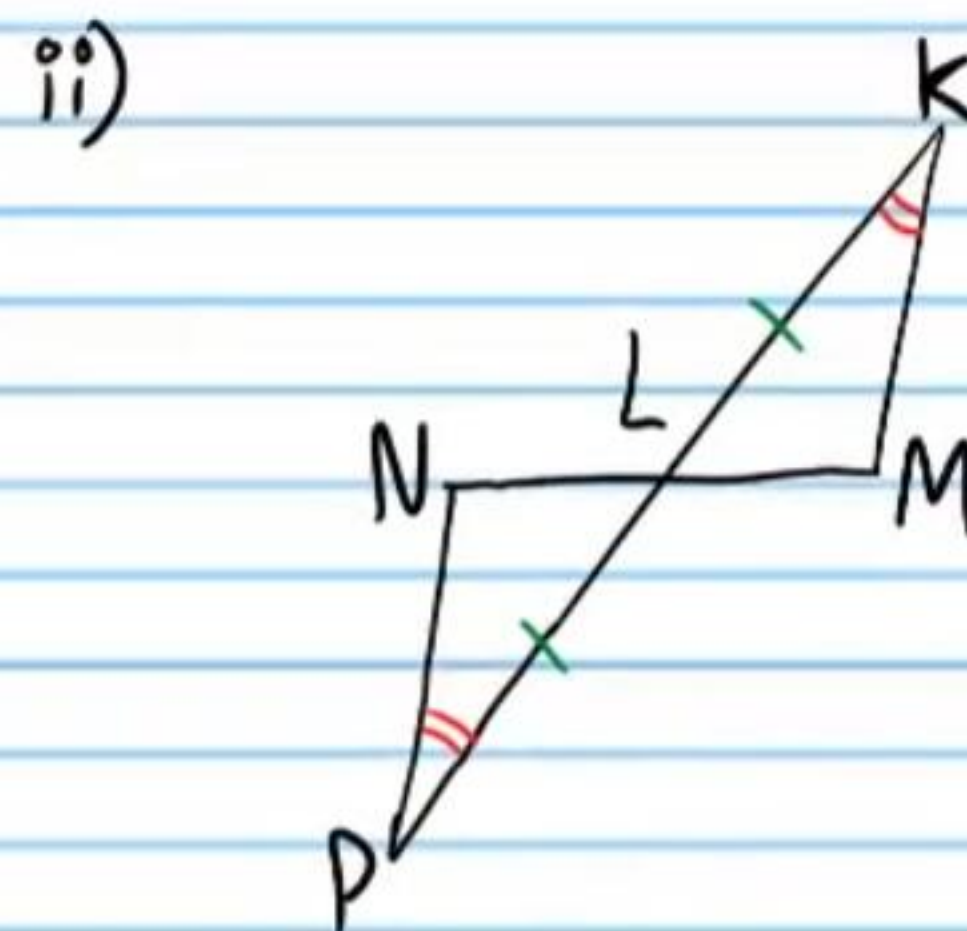
$\triangle EFG \cong$ _____



$\triangle ABC \cong$ _____

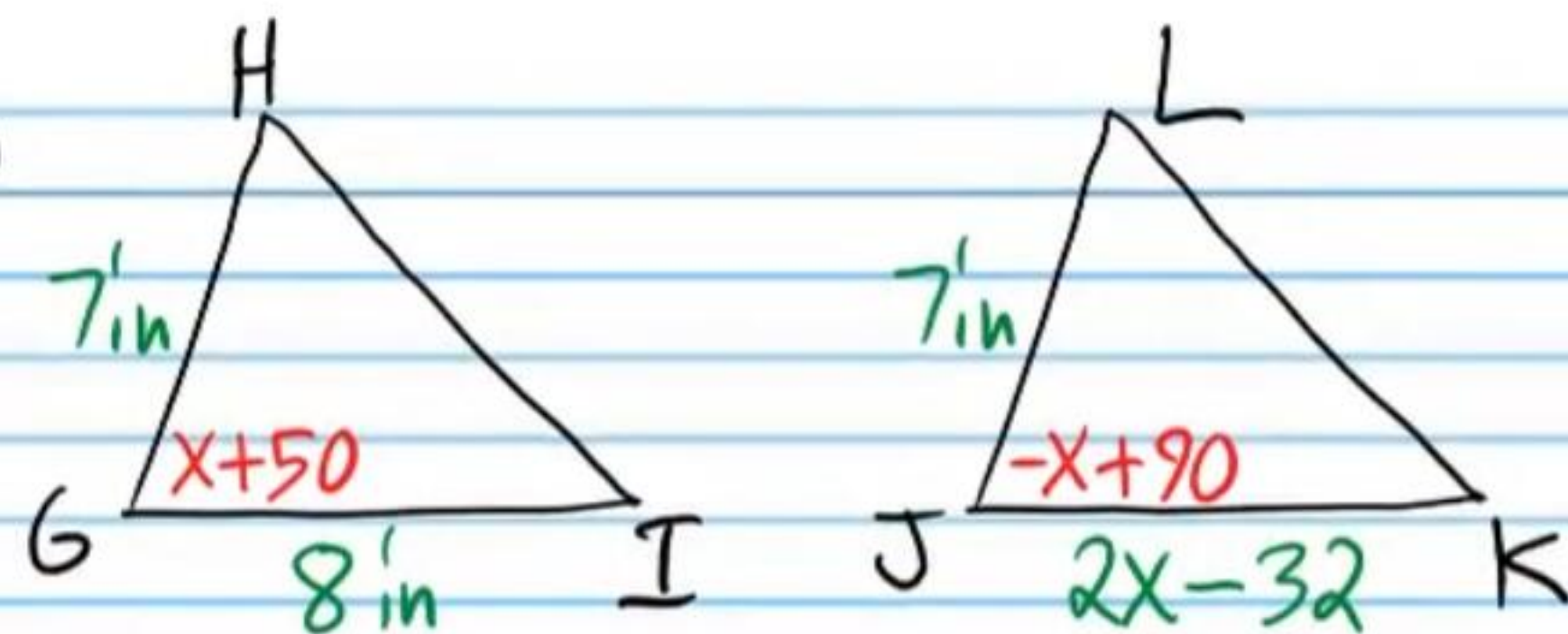


$\triangle ABC \cong$ _____



$\triangle KML \cong$ _____

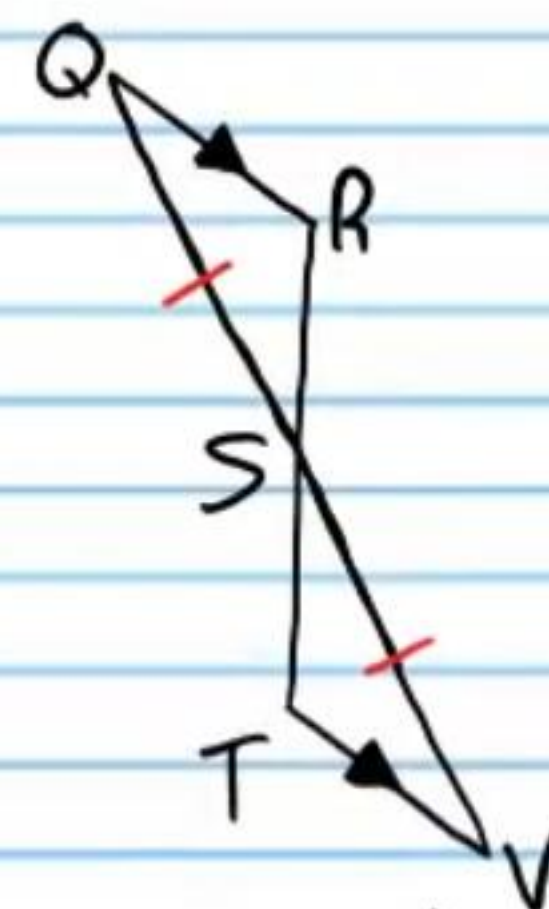
11 i) $x = 20$



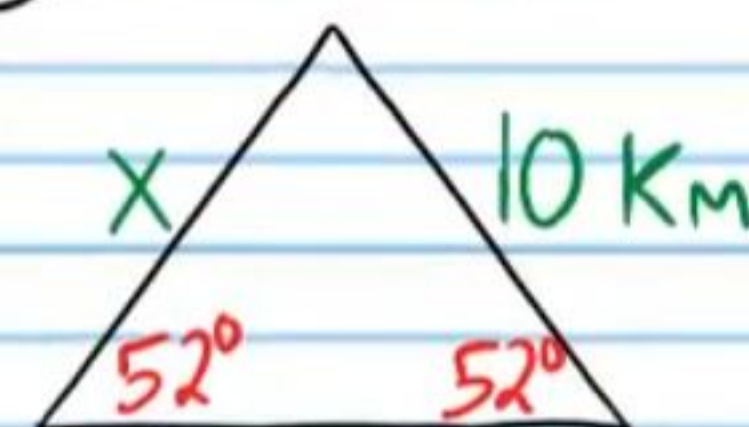
$\triangle GHI \cong$ _____

ii) $\overline{QR} \parallel \overline{TV}$.

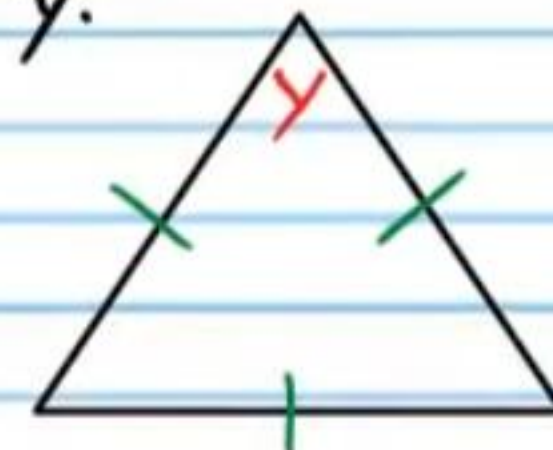
$\triangle QRS \cong$ _____



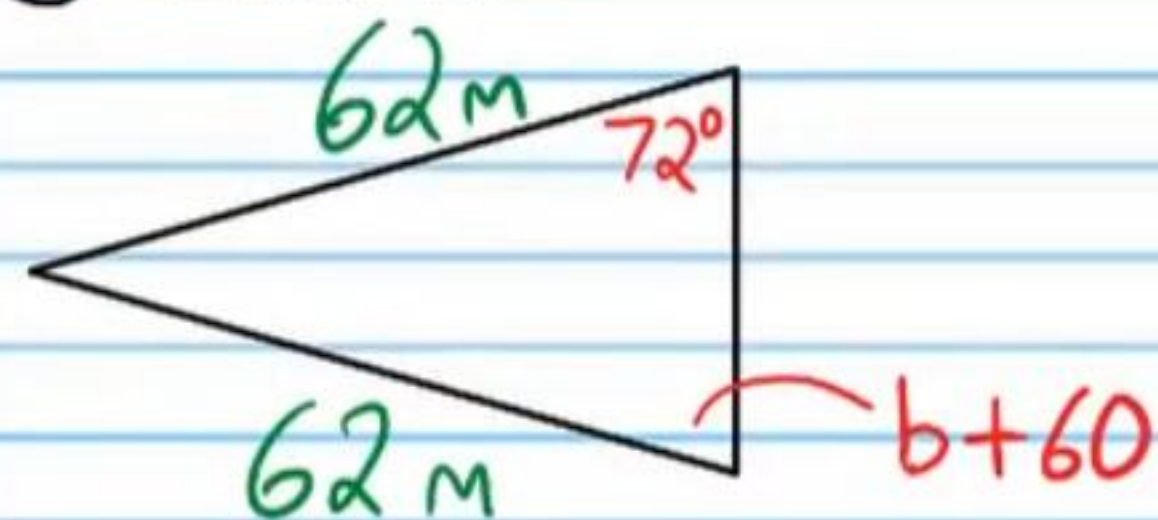
12 Find X.



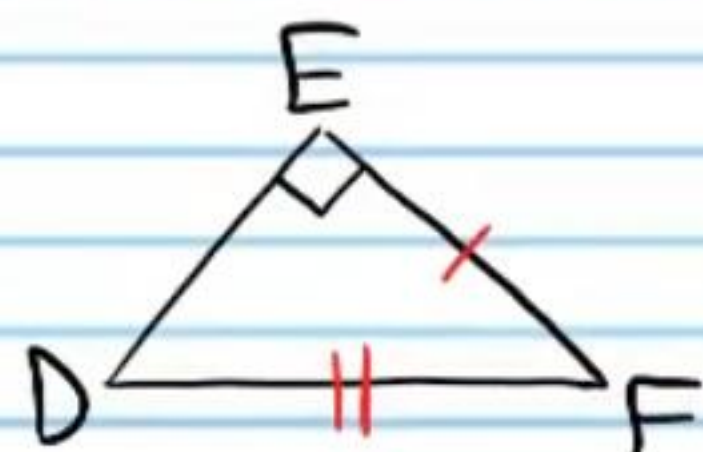
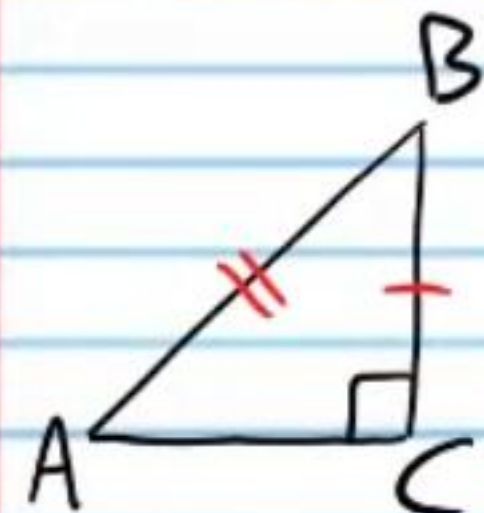
13 Find y.



⑭ Find b .

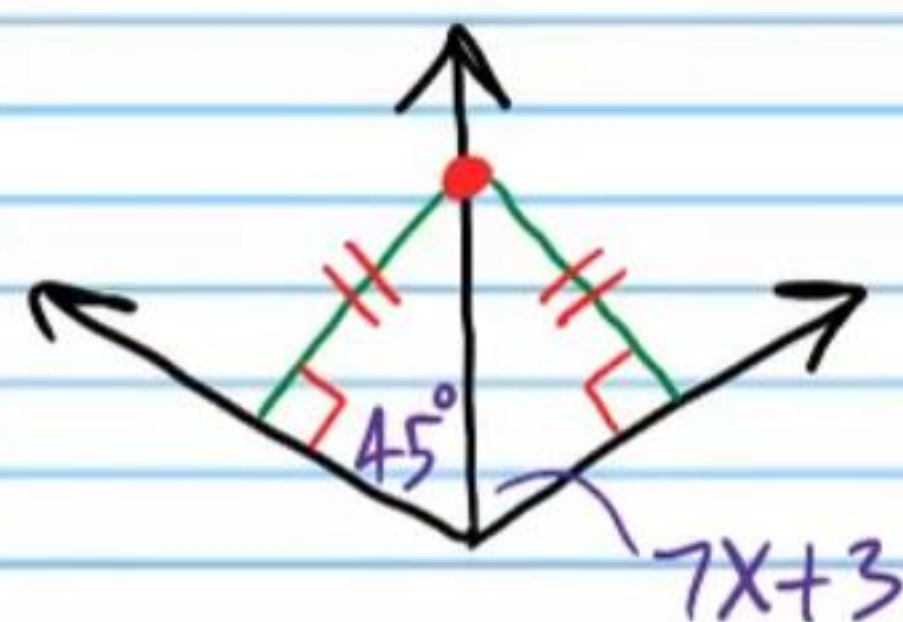


⑮ Are the triangles below congruent? If so, write a proper congruence statement and the theorem or postulate used.

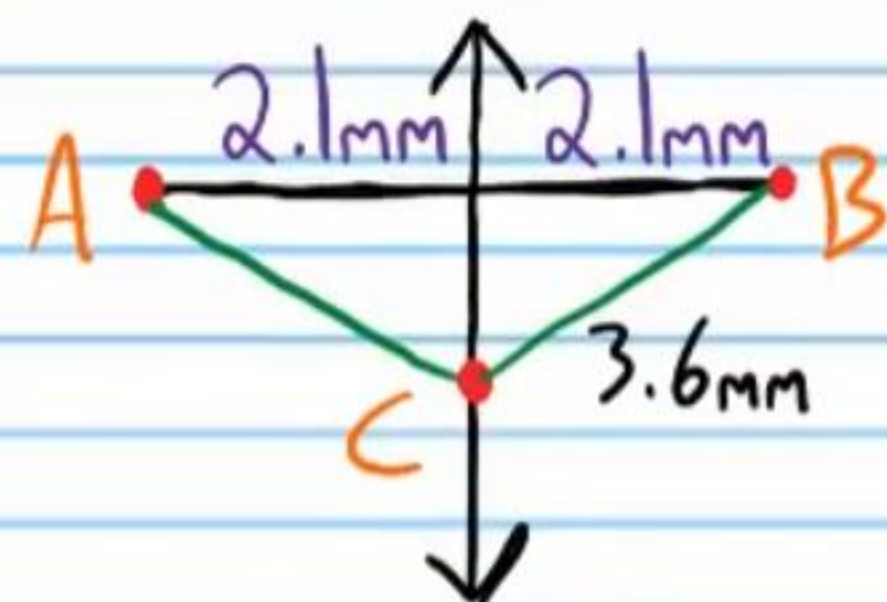


$\triangle ABC \cong$ _____

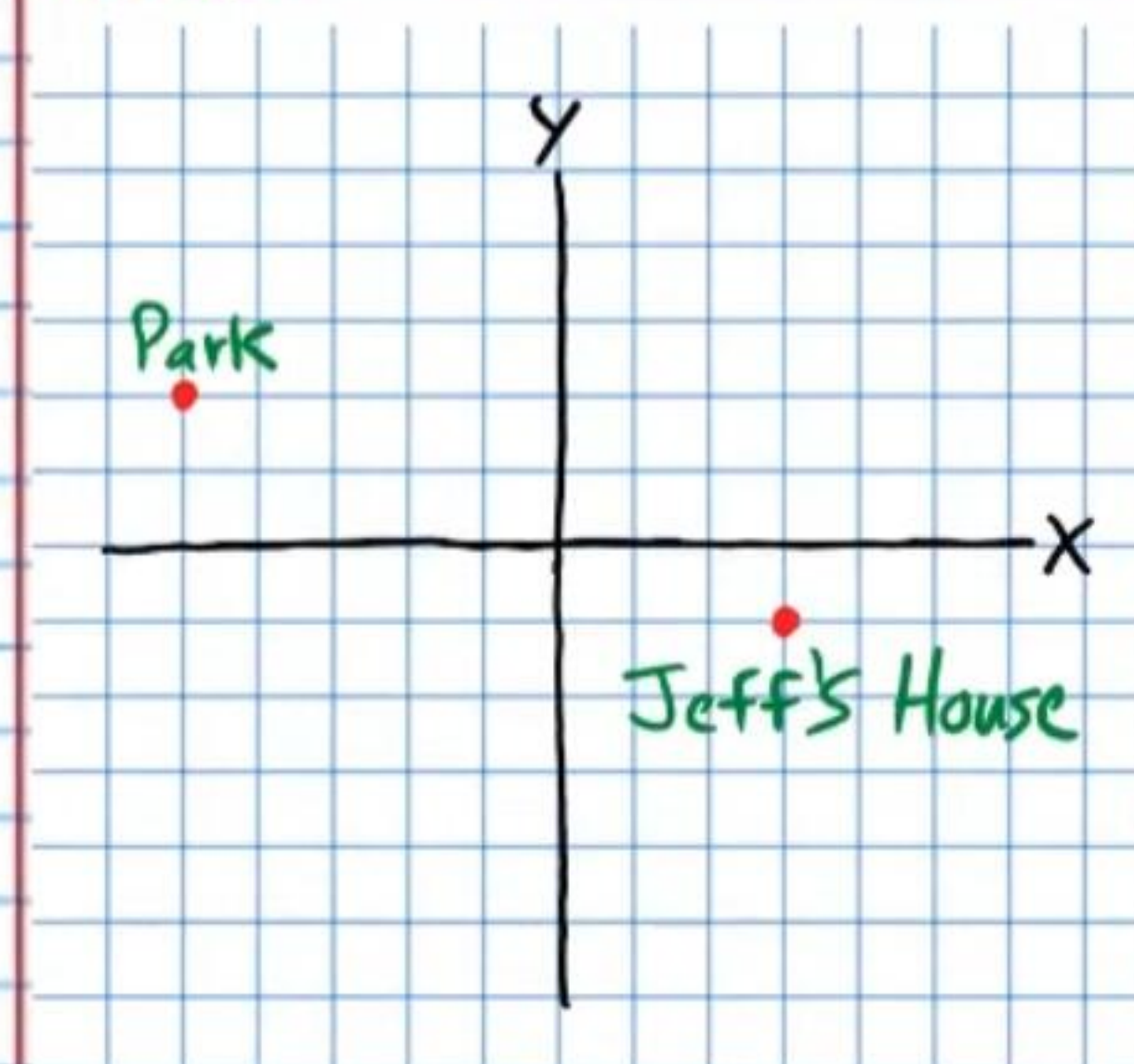
⑯ Find x .



⑰ Find AC .

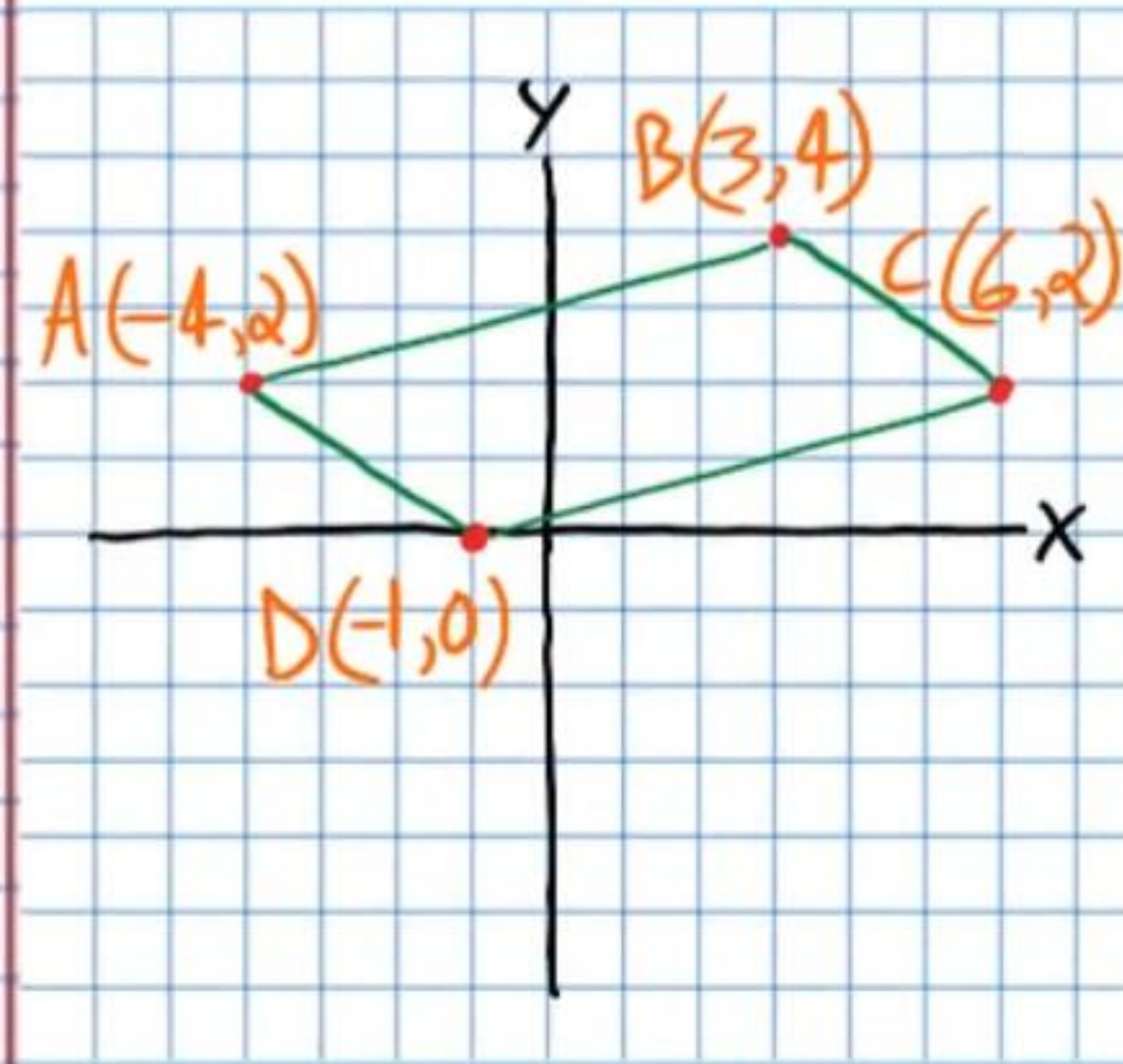


⑱ Find the distance between Jeff's house and the park. Each unit on the grid represents two miles.

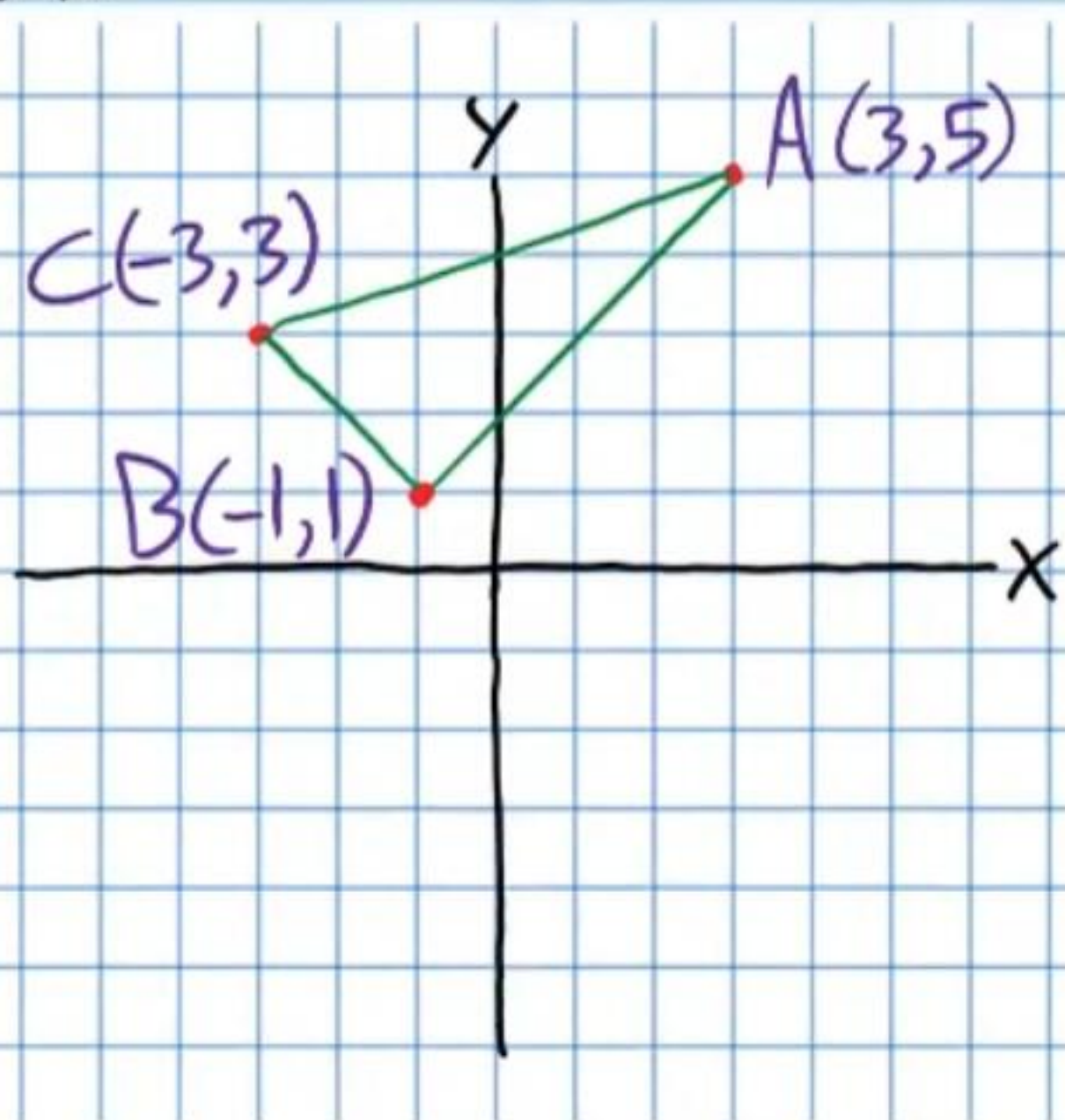


⑪ If $(2, -4)$ is the midpoint between points A and B, and the coordinates of A are $(0, 7)$, find the coordinates of point B.

⑫ Show that the quadrilateral below is a parallelogram.



②1 Show that the midsegment connecting $\overline{CA}$ and $\overline{AB}$ is half the length of $\overline{CB}$ and parallel to $\overline{CB}$ as well.



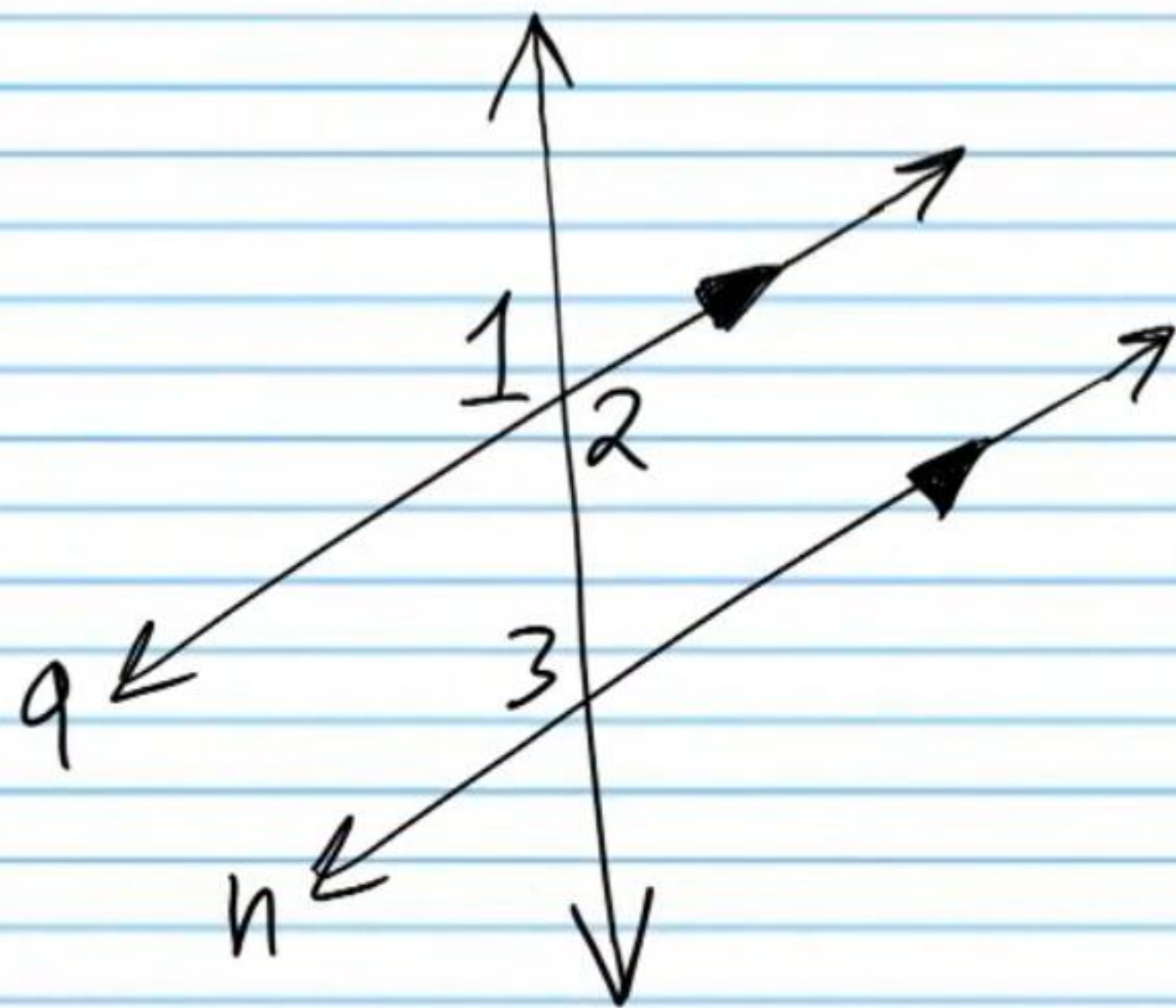
Show that the following statements are true using two-column proofs. You might not use the same number of steps written in Roman numerals. You may use more or less.

②2 If $\overline{DB}$ bisects $\angle B$ and $\angle D$ below, then $AD = DC$.



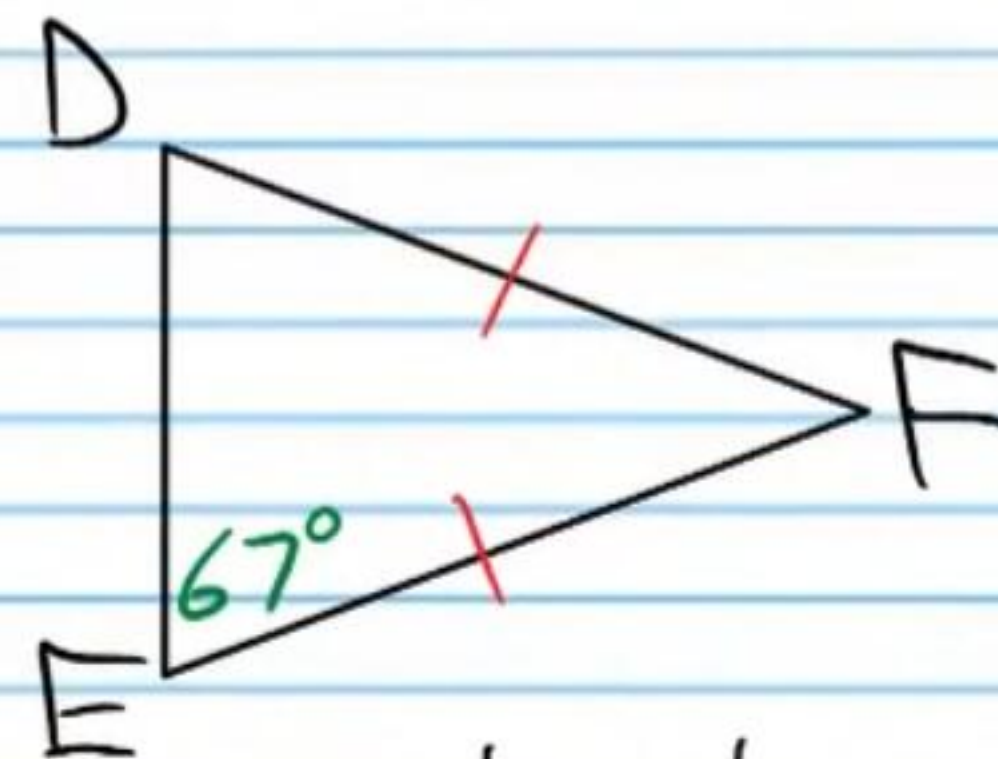
Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	

②③ If $q \parallel n$ below, then $\angle 2 \cong \angle 3$. Note: This is the proof of the Alternate Interior Angles Theorem. *You must use the Corresponding Angles Postulate and the Vertical Angles Theorem.*



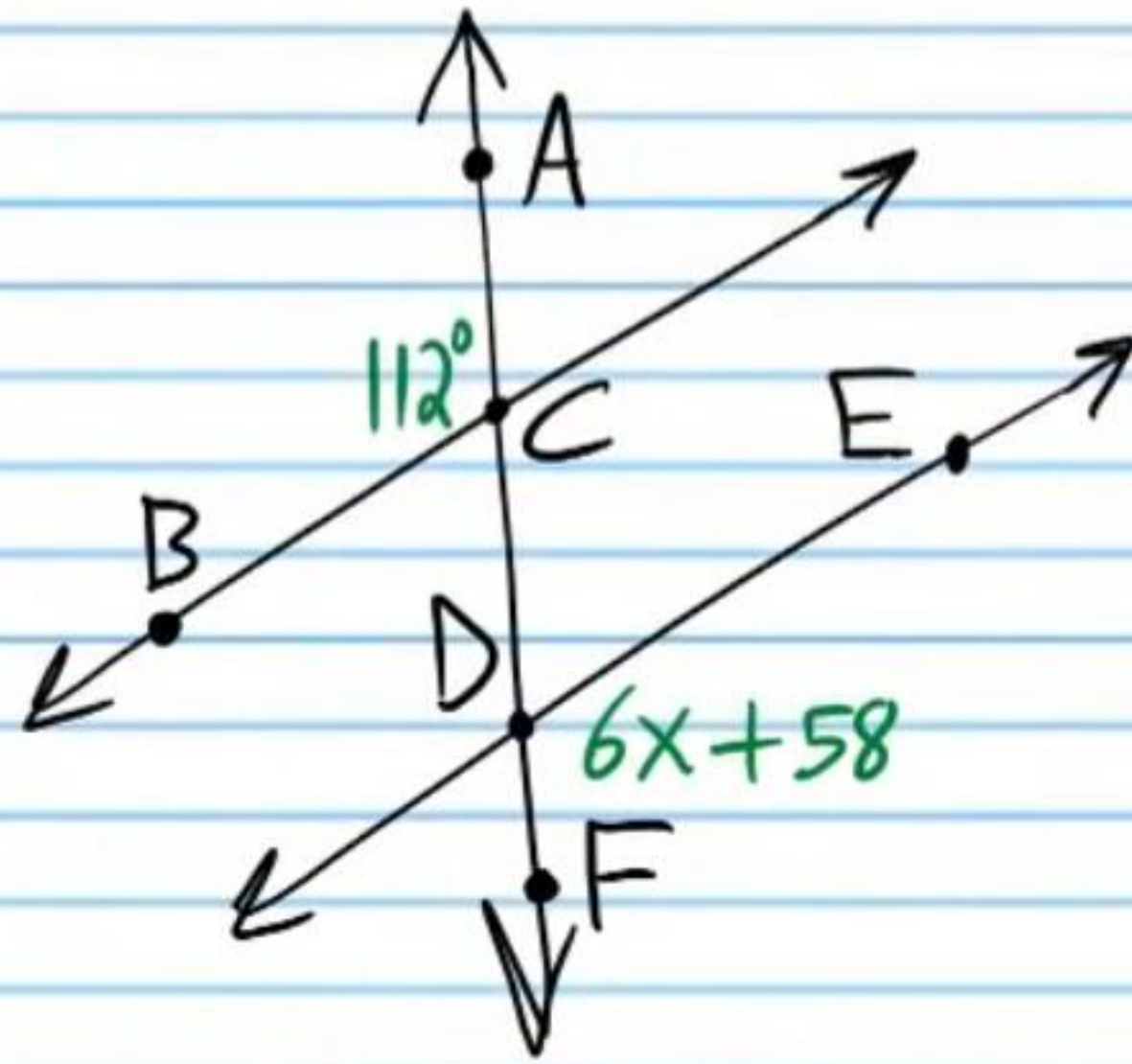
Statement	Reason
I)	
II)	
III)	
IV)	

②④ If $\overline{DF} \cong \overline{EF}$ below, then $m\angle D = 67^\circ$. You must use the Isosceles Triangle Theorem.



Statement	Reason
I)	
II)	
III)	
IV)	
V)	

Q5) If $m\angle ACB = 112^\circ$ and $x = 9$, then $\overleftrightarrow{CB} \parallel \overleftrightarrow{ED}$. Use the Alternate Exterior Angles Converse Theorem.



Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	
VII)	

Geometry Test II Answers

$$\begin{array}{lll} \textcircled{1} \quad x = 100^\circ & a = 100^\circ & d = 100^\circ \\ & b = 100^\circ & \\ & c = 80^\circ & \end{array}$$
$$\begin{array}{l} y = 80^\circ \\ z = 80^\circ \end{array}$$

$$\textcircled{2} \quad x = 37, \text{ CIAT}$$

$\textcircled{3}$ i) Scalene

ii) Adjacent angles

$$\textcircled{4} \quad z = 23$$

$$\textcircled{5} \quad y = 44^\circ$$

$$\textcircled{6} \quad 12 \text{ ft}$$

$$\textcircled{7} \quad x = \sqrt{39}$$

$$\textcircled{8} \quad a = \sqrt{11}$$

$\textcircled{9}$ i) $\triangle EFG \cong \triangle JIH$ by AAS.

ii) $\triangle ABC \cong \triangle ADC$ by SSS

$\textcircled{10}$ i) Not enough information.

ii) $\triangle KML \cong \triangle PNL$ by ASA

$\textcircled{11}$ i) $\triangle GHI \cong \triangle JLK$ by SAS

ii) $\triangle QRS \cong \triangle VTS$ by ASA or AAS

$$\textcircled{12} \quad x = 10 \text{ km}$$

$$\textcircled{13} \quad y = 60^\circ$$

$$\textcircled{14} \quad b = 12$$

$\textcircled{15}$ Yes. $\triangle ABC \cong \triangle DFE$ by HL

$$\textcircled{16} \quad x = 6$$

$\textcircled{17}$ Not enough information. We know that the line bisects $\overline{AB}$, but we do not know that it is perpendicular to $\overline{AB}$. Therefore, we cannot use the Perpendicular Bisector Theorem.

$$\textcircled{18} \quad 2\sqrt{73} \text{ mi}$$

$$\textcircled{19} \quad B(4, -15)$$

$$\textcircled{20} \text{ I) } m_{AB} = \frac{2}{7} \text{ and } m_{DC} = \frac{2}{7}$$

$$\text{II) } m_{AD} = -\frac{2}{3} \text{ and } m_{BC} = -\frac{2}{3}.$$

Therefore, the quadrilateral is a parallelogram.

$$\textcircled{21} \text{ I) } m_{AC}(0, 4) \text{ and } m_{AB}(1, 3)$$

$$m_{mid} = -1 \text{ and } m_{CB} = -1$$

$$\text{II) } D_m = \sqrt{2} \text{ and } D_{BC} = 2\sqrt{2}$$

* Your proofs may be slightly different from those given below, but they could still be correct if they make sense.

(22)

Statement	Reason
I) $\overline{DB}$ bisects $\angle B$ and $\angle D$.	Given.
II) $\angle ABD \cong \angle CBD$ and $\angle ADB \cong \angle CDB$.	Def. of $\angle$ bisector.
III) $\overline{BD} \cong \overline{BD}$	Ref. Prop of $\cong$
IV) $\triangle ABD \cong \triangle CBD$	ASA
V) $\overline{AD} \cong \overline{DC}$	CPCTC
VI) $AD = DC$	Def of $\cong$ segs.

(23)

Statement	Reason
I) $q \parallel n$	Given.
II) $\angle 1 \cong \angle 3$	CAP
III) $\angle 1 \cong \angle 2$	Vert. $\angle$'s Thm.
IV) $\angle 2 \cong \angle 3$	Trans. Prop of $\cong$

(24)

Statement	Reason
I) $\overline{DF} \cong \overline{EF}$	Given.
II) $\angle D \cong \angle E$	Isos. $\triangle$ Thm
III) $m\angle D = m\angle E$	Def. of $\cong \angle$'s.
IV) $m\angle E = 67^\circ$	Given (diagram)
V) $m\angle D = 67^\circ$	Trans. Prop of $=$.

(25)

Statement	Reason
I) $x = 9$	Given.
II) $m\angle EDF = 6x + 58$	Given (diagram)
III) $m\angle EDF = 6(9) + 58$	Subs. Prop of $=$.
IV) $m\angle EDF = 112^\circ$	Simplification.
V) $m\angle ACB = 112^\circ$	Given.
VI) $m\angle EDF = m\angle ACB$	Trans. Prop of $=$.
VII) $\angle EDF \cong \angle ACB$	Def. of $\cong \angle$'s
VIII) $\overleftrightarrow{CB} \parallel \overleftrightarrow{ED}$	AEACT